This is CS50

CS50's Introduction to Computer Science

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David J. Malan (https://cs.harvard.edu/malan/)
malan@harvard.edu
(https://www.facebook.com/dmalan) (https://github.com/dmalan)
(https://www.instagram.com/davidjmalan/) (https://www.linkedin.com/in/malan/)
(https://www.reddit.com/user/davidjmalan) (https://x.com/davidjmalan)

The End

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Recap

- Over these past ten weeks, you have been drinking from the proverbial firehose.
- During this course, you learned to program in many different languages; indeed, our great hope is that you *learned how to program* in them all, regardless of the language.
- Consider the various topics we discussed over the last ten weeks: You have come so far and have gained so many skills.
- You should measure your success as the delta between your skill at the start of the course and your skill now, at the end of the course.
- Further, we hope that you *learned how to solve problems* above all else, how to take input, process it through an algorithm, and produce some output. To that point, we played a game where we drew a picture using verbal instructions.
- Notice how some verbal instructions were *higher level* and others were *lower level*, like the different programming languages utilized in this course.

- Notice how some instructions were more specific than others. Remember times when you had to refine your code to do exactly as you wanted.
- Also, notice how some instructions were abstracted away, like *draw arms*. This is similar to how we abstracted away code into functions.
- Looking at all the weeks of the course, you gathered tools and built skills. Above all, though, you learned how to teach yourself and leverage learning resources.

Final Project

- Your final project is your opportunity to build a project, using these skills, that is of your specification and design. We hope you will join us for the CS50 Hackathon (https://cs50.harvard.edu/college/2025/fall/hackathon/), an epic all-nighter wherein we work together on our final projects.
- Further, you should notice in the final project specification the opportunity to leverage AI-based tools outside the course's own (not permitted in any other problem set).
- For those of you on campus, we hope that you will attend the CS50 Fair to showcase your final project.

Looking Ahead

- As you journey from the work of this course to the world outside CS50, you may want to take a number of steps to prepare and adventure out.
- Journeying on, download (https://code.visualstudio.com/) and learn (https://code.visualstudio.com/docs) about VS Code. Cursor (https://cursor.com) and Windsurf (https://windsurf.com) are other popular text editors.
- To be able to execute commands on the terminal, much like you did on CS50.dev (https://cs50.dev), install command-line tools on your Mac (https://developer.apple.com/xcode/) or PC (https://learn.microsoft.com/en-us/windows/wsl/about).
- Learn more about Git (https://youtu.be/MJUJ4wbFm_A).
- Host a website using GitHub (https://pages.github.com/).
- If applicable, host a web app using AWS (https://aws.amazon.com/education/awseducate/), Azure (https://azure.microsoft.com/en-us/free/students/), Google Cloud (https://cloud.google.com/edu/students), Cloudflare (https://www.cloudflare.com/students), Heroku (https://heroku.com), or Vercel (https://vercel.com).
- Ask questions in relevant online communities.
- Ask questions using AI-based tools like ChatGPT (https://chat.openai.com/), Claude (https://claude.ai), Gemini (https://gemini.google.com), GitHub Copilot

(https://github.com/features/copilot), Codex (https://openai.com/codex), or v0 (https://v0.app).

- Take any of our other CS50 courses (https://edX.org/cs50).
- Join one of our many communities (https://cs50.harvard.edu/communities).

Thank You!

- Thank you for engaging in this course!
- Many people made this course possible: Thank you for making this course possible and for supporting us so well!

Summing Up

In this lesson, we reflected on your journey through this course. Specifically, we encourage you to…

- Take the programming skills you learned in this course to solve problems in the world.
- Take the new skills you acquired in this course and continue your learning!
- Finish strong and submit your final project.
- Quack! (https://youtu.be/DLzxrzFCyOs?si=PC0oK7Pv_NkO4FmX)

This was CS50!